

Mathematical Problems in Deep Learning: Homeworks #1 Solutions

1. **Probability Bound on Query Differences:** Consider a data $D = (d_1, d_2, \dots, d_n)$ which is a n dimensional binary vector and another data $\tilde{D} = (\tilde{d}_1, \tilde{d}_2, \dots, \tilde{d}_n)$. The difference between datasets is measured by

$$l(D, \tilde{D}) = \sum_{i=1}^n \mathbf{1}(d_i \neq \tilde{d}_i).$$

Define $q_{s_j}(D)$ as the response obtained by querying a subset of indices s_j . Let

$$B = \{\tilde{D} \mid l(D, \tilde{D}) \geq 256\alpha^2 n^2\}.$$

Prove that for $D_0 \in B$

$$\Pr(|q_{s_j}(D_0) - q_{s_j}(D)| \leq 4\alpha n) \leq \frac{9}{10}.$$

Solution:

Assume that two data $D = (d_1, \dots, d_n)$ and $D_0 = (d_{0,1}, \dots, d_{0,n})$ differ in at least

$$k \geq 256\alpha^2 n^2$$

positions. Let $F = (f_1, \dots, f_n)$ be a n dimensional binary vector where $f_i = 1$ if and only if $i \in s_j$. Then,

$$\begin{aligned} X &= |q_{s_j}(D_0) - q_{s_j}(D)| \\ &= |F \cdot (D_0 - D)|. \end{aligned}$$

Note that $D_0 - D \in \{-1, 0, 1\}^n$ is an n dimensional vector at least $k = 256\alpha^2 n^2$ of elements are nonzero. Then, for large enough n , the $F \cdot (D_0 - D)$ behaves like Gaussian with variance σ^2 at least $k/4 = 64\alpha^2 n^2$. This is because

$$\begin{aligned} \text{Var}(F \cdot (D_0 - D)) &= \sum_{i=1}^n \text{Var}(f_i(d_{0,i} - d_i)) \\ &= \sum_{d_{0,i} \neq d_i} \frac{1}{4} \\ &\geq k/4 = 64\alpha^2 n^2. \end{aligned}$$

Finally, for $Z \sim \mathcal{N}(0, (8\alpha n)^2)$

$$\begin{aligned} \Pr[|F \cdot (D_0 - D)| \leq 4\alpha n] &\leq \Pr[|Z| \leq 4\alpha n] \\ &= \Pr[|Z| \leq \sigma/2] \\ &\leq \frac{9}{10} \end{aligned}$$

2. **Differential Privacy Amplification via Subsampling:** Suppose we have an algorithm $M : X^m \rightarrow Y$ which is ϵ -DP. Consider the following algorithm $M' : X^n \rightarrow Y$, where $n > m$. When run on an input $X \in X^n$, it chooses a random subset of the input $X' \in X^m$ of size m , and outputs $M(X')$. Let $q = m/n$.

Let $X, X' \in X^n$ be neighboring datasets. Without loss of generality, assume that:

$$X = (x_1, x_2, \dots, x_n), \quad X' = (x'_1, x_2, \dots, x_n).$$

Define the sampling process: Let $I = (I_1, \dots, I_m)$ be a random vector of size m , uniformly sampled (without replacement) from $\{1, \dots, n\}$. Define the sampling function:

$$s(X; I) = (x_{I_1}, \dots, x_{I_m}).$$

The new mechanism M' operates as: $M'(X) = M(s(X; I))$.

Prove the following inequality:

$$\Pr(M(s(X; I)) \in T \mid 1 \in I) \leq e^\epsilon \Pr(M(s(X; I)) \in T \mid 1 \notin I).$$

For a pair of neighboring subsets $I_1, I_2 \subset \{1, \dots, n\}$ of size m **Solution:** where $1 \in I_1$ and $1 \notin I_2$, we say $I_1 \sim_m^{(1)} I_2$. In other words, I_2 is replacing element '1' of I_1 with other index. Then, we have

$$\Pr[M(s(X; I_1)) \in T] \leq e^\epsilon \Pr[M(s(X; I_2)) \in T]. \quad (1)$$

Consider all possible such pair of (I_1, I_2) and sum them up, then

$$\sum_{I_1 \sim_m^{(1)} I_2} \Pr[M(s(X; I_1)) \in T] \leq e^\epsilon \sum_{I_1 \sim_m^{(1)} I_2} \Pr[M(s(X; I_2)) \in T]. \quad (2)$$

For each subset of size m set $1 \in I_1$, there are $n - m$ possible I_2 's since the element '1' can be replaced by elements in I_1^c . For each subset of size m set $1 \notin I_2$, there are m possible I_1 's since every element can be replaced by element '1'.

$$\sum_{I_1: |I_1|=m, 1 \in I_1} \Pr[M(s(X; I_1)) \in T] m \leq e^\epsilon \sum_{I_2: |I_2|=m, 1 \notin I_2} \Pr[M(s(X; I_2)) \in T] (n - m). \quad (3)$$

Thus, we have

$$\Pr[M(s(X; I_1)) \in T, 1 \in I_1] m \leq e^\epsilon \Pr[M(s(X; I_2)) \in T, 1 \notin I_2] (n - m). \quad (4)$$

This is equivalent to

$$\Pr[M(s(X; I_1)) \in T \mid 1 \in I_1] \frac{1}{\binom{n-1}{m-1}} m \leq e^\epsilon \Pr[M(s(X; I_2)) \in T \mid 1 \notin I_2] \frac{1}{\binom{n-1}{m}} (n - m), \quad (5)$$

and finally we get

$$\Pr[M(s(X; I_1)) \in T \mid 1 \in I_1] \leq e^\epsilon \Pr[M(s(X; I_2)) \in T \mid 1 \notin I_2]. \quad (6)$$

3. **Sensitivity of K-Means Algorithm:** Consider a K-means clustering algorithm where d -dimensional data points $X_1, \dots, X_n \in U$, where $U = \{x \in \mathbb{R}^d \mid \|x\|_1 \leq 1\}$. Given the centers $C_1, \dots, C_k$, the mechanism f first group X_j 's by

$$G_i = \{X_j : d(X_j, C_i) \leq d(X_j, C_m) \text{ for all } m \neq i\}$$

Then, the mechanism f outputs $f(C_1, \dots, C_k) = (C'_1, \dots, C'_k)$ where

$$C'_i = \frac{1}{|G_i|} \sum_{X_j \in G_i} X_j.$$

Determine the sensitivity of f .

Solution:

Let $x, x' \in \mathbb{R}^d$ be two data points such that $\|x\|_1 \leq 1$ and $\|x'\|_1 \leq 1$. Consider the worst-case scenario where the data set D and its neighboring data set D' differ in exactly one data point, and this data point belongs to a cluster that contains only that point.

In this case, the cluster mean in $f(D)$ is simply x , and the cluster mean in $f(D')$ is x' . Therefore, the change in output is:

$$\|f(D) - f(D')\|_1 = \|x - x'\|_1 \leq \|x\|_1 + \|x'\|_1 \leq 2.$$

Thus, the maximum possible change in the output (i.e., the global sensitivity) is 2, which occurs when a singleton cluster's data point is replaced by another point at L_1 distance 2.

Hence, the global sensitivity of f is $\boxed{2}$.

4. **Proof of Additive δ -DP:** We aim to show that M satisfies additive δ -Differential Privacy (δ -DP), which is defined as:

$$|\Pr(M(D) \in S) - \Pr(M(D') \in S)| \leq \delta,$$

for any neighboring datasets D and D' differing by at most one element and for any subset S of the output space.

- (a) Consider a mechanism $M(X)$ defined for a dataset $X = (X_1, \dots, X_n)$ as follows:

$$M(X) = \begin{cases} X & \text{with probability } \delta, \\ \perp & \text{with probability } 1 - \delta. \end{cases}$$

Show that M is additive δ -DP.

- (b) Consider a mechanism $M(x)$ defined for a dataset $X = (X_1, \dots, X_n)$ as follows:

$$M(X) = (Y_1, \dots, Y_n), \quad \text{where } Y_i = \begin{cases} X_i & \text{with probability } \delta, \\ \perp & \text{with probability } 1 - \delta. \end{cases}$$

Show that M is additive δ -DP.

Solution:

(a) Consider two neighboring datasets X and X' that differ in exactly one element. Let T be any subset of the output space, and define the event E as “the mechanism outputs either X or X' ”. The complement event $\bar{E}$ corresponds to the mechanism outputting $\perp$, which occurs with probability $1 - \delta$ regardless of the input.

Since the output is always $\perp$ under $\bar{E}$, the distributions of $M(X)$ and $M(X')$ are identical in this case. Therefore, we can write:

$$\Pr[M(X) \in T] = \Pr[M(X) \in T \mid \bar{E}] \cdot \Pr[\bar{E}] + \Pr[M(X) \in T \mid E] \cdot \Pr[E].$$

Because $\Pr[E] = \delta$, we have:

$$\Pr[M(X) \in T] = \Pr[M(X') \in T \mid \bar{E}] \cdot \Pr[\bar{E}] + \Pr[M(X) \in T \mid E] \cdot \delta.$$

Since $\Pr[M(X) \in T \mid E] \leq 1$, we get:

$$\Pr[M(X) \in T] \leq \Pr[M(X') \in T \mid \bar{E}] \cdot \Pr[\bar{E}] + \delta \leq \Pr[M(X') \in T] + \delta.$$

By symmetry, the same inequality holds with X and X' reversed. Hence:

$$|\Pr[M(X) \in T] - \Pr[M(X') \in T]| \leq \delta.$$

Therefore, the mechanism satisfies additive δ -differential privacy.

(b) Consider two neighboring datasets X and X' , which differ only at the i -th position. Let T be any subset of the output space (i.e., a set of possible outputs), and define the event E as “the i -th coordinate is revealed (i.e., not replaced by $\perp$)”. Then the complement $\bar{E}$ corresponds to the case where the i -th coordinate is replaced by $\perp$.

Conditioned on $\bar{E}$, the output distributions of $M(X)$ and $M(X')$ are identical, since all other coordinates are the same between X and X' . Therefore, we can write:

$$\Pr[M(X) \in T] = \Pr[M(X) \in T \mid \bar{E}] \cdot \Pr[\bar{E}] + \Pr[M(X) \in T \mid E] \cdot \Pr[E].$$

Note that $\Pr[E] = \delta$, and similarly, we get:

$$\Pr[M(X) \in T] \leq \Pr[M(X') \in T \mid \bar{E}] \cdot \Pr[\bar{E}] + 1 \cdot \delta.$$

But $\Pr[M(X') \in T \mid \bar{E}] \cdot \Pr[\bar{E}] = \Pr[M(X') \in T] - \Pr[M(X') \in T \mid E] \cdot \delta \leq \Pr[M(X') \in T]$. So we conclude:

$$\Pr[M(X) \in T] \leq \Pr[M(X') \in T] + \delta.$$

Thus, the mechanism satisfies additive δ -differential privacy.

5. **A different private algorithm:** Suppose that we wanted to answer a count query: $f(X) = \sum_{i=1}^n X_i$, where $X_i \in \{0, 1\}$. In class, we learned the Laplace mechanism: simply add Laplace noise with scale parameter $1/\epsilon$. But what if we do not have

access to Laplace noise? Suppose Z is a continuous uniform random variable, drawn uniformly from the interval $[-3/\epsilon, 3/\epsilon]$. Consider the statistic $\tilde{f}(X) = \sum_{i=1}^n X_i + Z$. Is $\tilde{f}$ $O(\epsilon)$ -differentially private? If yes, prove it, with the best constant you can give in the privacy guarantee. If no, explain why not.

Solution:

No it is not, let $X = (0, \dots, 0)$ and $X' = (1, 0, \dots, 0)$. Then, the density around $1 + 3/\epsilon$ of $\tilde{f}(X)$ would be zero, while the density around $1 + 3/\epsilon$ of $\tilde{f}(X')$ is strictly positive.

6. **Mechanisms:** Consider the following mechanisms M that takes a dataset $x \in [0, 1]^n$ and returns an estimate of the mean $\bar{x} = (\sum_{i=1}^n x_i)/n$.

(M1) $M_1(x) = [\bar{x} + Z]_0^1$, for $Z \sim \text{Lap}(2/n)$, where for real numbers y and $a \leq b$, $[y]_a^b$ denotes the “clamping” function:

$$[y]_a^b = \begin{cases} a & \text{if } y < a \\ y & \text{if } a \leq y \leq b \\ b & \text{if } y > b \end{cases}.$$

(M2)

$$M_2(x) = \begin{cases} 1 & \text{w.p. } \bar{x} \\ 0 & \text{w.p. } 1 - \bar{x} \end{cases}.$$

Which of the above mechanisms meet the definition of ϵ -differential privacy for a finite value of ϵ , and what is the smallest value of ϵ (possibly as a function of n) for which they do? As in class, here we are treating n as public knowledge (so it is not a privacy violation to reveal n), and working with the “change-one” definition of DP.

Solution:

- (a) The maximum probability difference happens when $\bar{x} = 1$ and the minimum $\bar{x} = 1 - 1/n$. Thus, for M_1 ,

$$\frac{\Pr(M_1(X) \in T)}{\Pr(M_1(X') \in T)} \leq \frac{f_X(1)}{f_X(1 - 1/n)} = e^{n/2(1/n)} = e^{1/2}.$$

where f_X is a pdf of Laplace distribution with parameter $2/n$. Thus, the minimum ϵ for M_1 is $1/2$.

- (b) Without loss of generality, it is enough to check the maximum ratio of probabilities at $M_2(X) = 1$.

$$\frac{\Pr(M_2(X) = 1)}{\Pr(M_2(X') = 1)} = \frac{\bar{X}}{\bar{X'}}.$$

This is unbounded since $\bar{X'}$ can be 0. Thus, it cannot be ϵ -DP for finite ϵ .

7. **Mean estimation with non-binary data:** In class, we saw how to estimate the mean of a dataset $\frac{1}{n} \sum_{i=1}^n X_i$ in the case when the X_i 's are binary. Here, we will see how to estimate the mean of a dataset when this may not be the case.

- (a) Suppose we only knew the $X_i \in \mathbb{R}$ were real numbers. Prove that, for all $t \geq 0$, there is no ϵ -DP algorithm $M : \mathbb{R}^n \rightarrow \mathbb{R}$ such that

$$\Pr(|M(X) - f(X)| \leq t) \geq 0.9,$$

where $\epsilon = 1$.

- (b) The previous problem showed that, in general, we can't privately estimate the mean of an unbounded dataset. Let's see how we can circumvent this issue. Give an algorithm $A_2 : \mathbb{R}^n \rightarrow \mathbb{R}$ with the following guarantees. The algorithm is ϵ -DP, for all possible datasets $(X_1, \dots, X_n) \in \mathbb{R}^n$. If all $X_i \in [-R, R]$, then there exists some constant $C > 0$ such that

$$\Pr(|A_2(X) - f(X)| \leq \frac{CR}{n\epsilon}) \geq 0.9.$$

The parameter R is known to the algorithm. Observe that this algorithm must always be private, but only needs to be accurate when the input dataset satisfies some additional properties

Solution:

- (a) Without any bound, we can set $X = (0, \dots, 0)$ and $X' = (2nt, 0, \dots, 0)$. Note that $f(X) = 0$ and $f(X') = 2t$. First, due to ϵ -DP, we have

$$\frac{\Pr(M(X) \in [-t, t])}{\Pr(M(X') \in [-t, t])} \leq e^\epsilon$$

which implies $\Pr(|M(X')| \leq t) \geq e^{-\epsilon} \Pr(|M(X)| \leq t)$.

Also, the following should be true for both X and X' ,

$$\begin{aligned} \Pr(|M(X)| \leq t) &\geq 0.9 \\ \Pr(|M(X') - 2t| \leq t) &\geq 0.9 \end{aligned}$$

This cannot be true since

$$\begin{aligned} 0.1 &\geq \Pr(|M(X') - 2t| > t) \\ &\geq \Pr(|M(X')| \leq t) \\ &\geq e^{-\epsilon} \Pr(|M(X)| \leq t) \\ &\geq e^{-\epsilon} 0.9. \end{aligned}$$

This is contradiction.

- (b) The sensitivity of f is $2R/n$. Thus, if we apply Laplace mechanism with Laplacian parameter $2R/n\epsilon$, we can achieve ϵ -DP. The probability of error is

$$\begin{aligned} Pr(|A_2(X) - f(X)| > CR/n\epsilon) &= Pr(|Y| > CR/n\epsilon) \\ &= e^{-C/2}. \end{aligned}$$

where Y is Laplace random variable of parameter $2R/n\epsilon$. Thus, we can set $C = 2 \log 10$ to get probability of error equals to 0.1. In other words,

$$Pr(|A_2(X) - f(X)| \leq (2 \log 10)R/n\epsilon) = 0.1$$